Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Nanoscale Electronic and Optoelectronic Devices for Advanced Information and Display Technologies			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		Electrical・Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Devices and Systems	Other Semiconductor Devices	Optical Application Devices	Optical Devices
Concept of Proposed Technology	In this project, the PI proposes to develop nanoscale electronic and optoelectronic devices in collaboration with small and medium-sized enterprises (SMEs) in Korea for applications in advanced computing, communication, and display technologies. Specifically, we will develop multi-channel logic transistors, plasmon-enhanced photodetectors, and flexible light-emitting diodes (LEDs) based on two-dimensional (2D) materials. High-performance multi-channel logic transistors can provide both computation and logic control for sensors, displays, and biomedical devices. Highly sensitive plasmon-enhanced photodetectors can enable light detection across a broad spectral range, from the far-infrared (IR) to the visible. Flexible LEDs can enable displays on curved surfaces and foldable mobile phones, as well as provide illumination for medical implant devices. These logic transistors, photodetectors, and LEDs are all based on a common material platform, 2D materials, which enables their seamless integration into a single system. The resulting device			

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	platform has broad potential applications in advanced computing and communication systems, including on-chip optical interconnects, smart sensor nodes with embedded processing, and next-generation display and optoelectronic technologies. This work establishes a foundation for future integrated electronic and optoelectronic systems relevant to semiconductor, communication, and display applications. In addition, these flexible devices hold promise for biomedical applications, such as medical implants and wearable health monitors.
Rationale for Joint R&D Project	The proposed joint R&D project is driven by the growing need to translate emerging nanoscale electronic and optoelectronic device concepts into scalable, application-relevant technologies. While academic research is well suited to exploring new materials, device physics, and proof-of-concept demonstrations, advancing these innovations toward practical semiconductor, communication, and display applications requires complementary expertise in process integration, manufacturability, and system-level considerations. Collaboration with SMEs is essential to bridge this gap. SMEs provide critical expertise in semiconductor fabrication, next-generation displays, and application-driven design that are often not available within a university setting. Their involvement ensures that material selection, device architectures, and fabrication approaches are aligned with realistic manufacturing constraints and market requirements. This joint R&D program serves as a vital link between academic research and industrial development, fostering innovation in semiconductor and display technologies while enabling a synergistic approach in which fundamental research is guided by practical implementation pathways. This collaboration will accelerate technology maturation and reduce development risks. This international partnership will strengthen the leadership positions of both Korean enterprises and UIUC in microelectronics and nanotechnology.
Keywords	1. Field-effect transistor 2. Photodetector 3. Light emitting diode 4. Two-dimensional material 5. Flexible device

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (6)

R&D Objectives

The objectives of this R&D project are to develop high-performance logic transistors, plasmon-enhanced photodetectors, and flexible LEDs for advanced computing, sensing, and display applications. The PI brings extensive experience in electronic and photonic devices, which will be leveraged in this project. Specifically, we will undertake the following three research tasks.

1. **Development of High-Performance Logic Transistors Based on Silicon Nanosheets and 2D Semiconductors.**

Many AI algorithms and signal-processing tasks are computationally intensive and require substantial processing power, driving the need for high-performance logic transistors. Beyond FinFETs, silicon nanosheet transistors have emerged as the prevalent architecture in advanced CMOS technologies due to their excellent electrostatic control, high drive current, and tunable channel width. However, as the channel length scales below the ten-nanometer regime, device performance deteriorates significantly because of short-channel effects, mobility degradation, and high contact resistance.

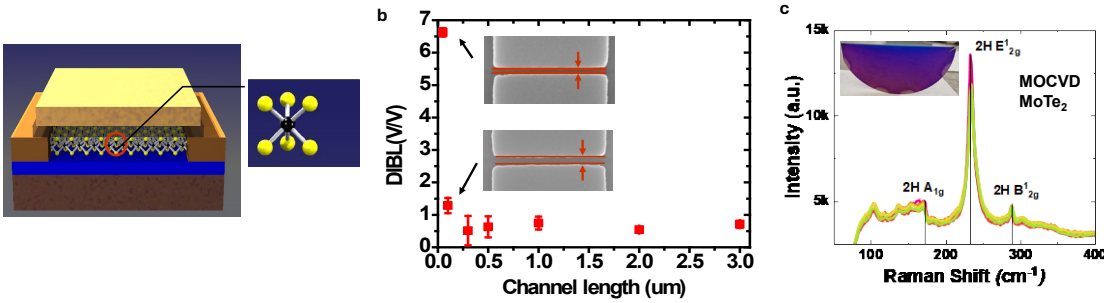

Figure 1. Preliminary results: logic transistors based on 2D materials. (a) Schematic of a MOSFET with monolayer MoS2 [1]. (b) Drain-induced barrier lowering (DIBL) of a MOSFET with CVD-grown MoS_2 at various channel lengths [1]. The inset shows the SEM image of transistors with 32 and 80 nm channel lengths. (c) Raman spectra measured at six different locations across a 4-inch wafer with $MoTe_2$ grown by MOCVD [7]. The inset shows an optical image of the 4-inch wafer with uniform $MoTe_2$ coverage.

The PI has demonstrated aggressively scaled MoS_2 logic transistors with suppressed short-channel effects [1]. The device structure is illustrated in Fig. 1a, while the drain-induced barrier lowering (DIBL) and SEM images of the MoS_2 transistors are shown in Fig. 1b. A minimum scalable channel length of 7 nm is extrapolated for a monolayer MoS_2 transistor with a 3 nm HfO_2 gate insulator. In addition, the PI's group has developed wafer-scale synthesis of high-quality transition metal dichalcogenides (TMDs) [2-7], which provides an excellent material platform for large-scale fabrication of 2D nanosheet transistors and circuits. The inset of Fig. 1c shows an optical image of a 4-inch $MoTe_2$ wafer grown by MOCVD [7]. Raman spectra taken at six different locations across the 4-inch $MoTe_2$ wafer are nearly identical, confirming the excellent uniformity of the $MoTe_2$ film [7].

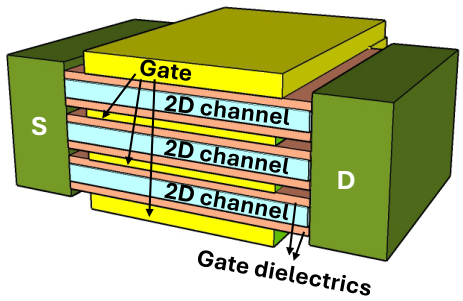

Figure 2. Schematic of proposed logic transistors with multiple 2D channels.

In this project, we propose to develop complementary multi-channel transistors using ultra-thin silicon nanosheet and 2D semiconductors, as illustrated in Fig. 2. Ultra-thin silicon offers the advantages of large-scale, high-quality materials and mature process technology, while 2D semiconductors provide atomically thin channel with dangling-bond-free surfaces. Stacking multiple channels will enhance the on-state current without consuming additional space on the chip.

2. Investigation of Plasmon-Enhanced Photodetectors

2D materials exhibit exceptional electronic and optical properties, including tunable bandgaps, broadband light absorption, and ultrafast response speeds, making them highly attractive for photodetectors. However, their atomically thin nature limits light absorption, posing a significant challenge to achieving high responsivity.

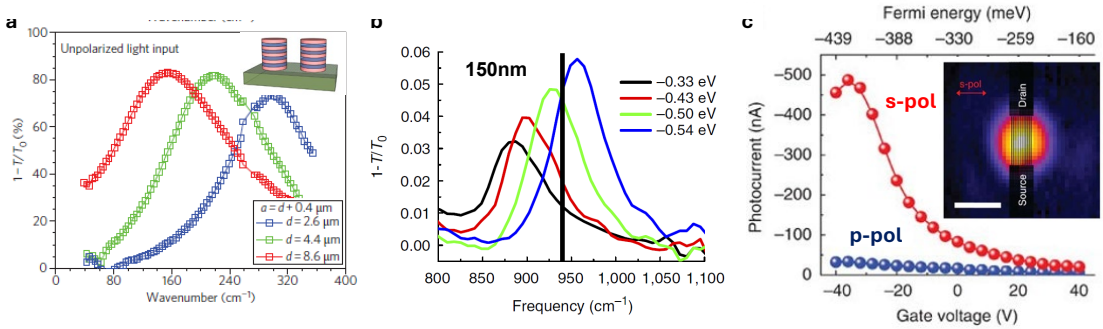

Figure 3. Preliminary results: plasmon-enhanced photodetectors. (a) Extinction spectra of graphene nanodisk arrays fabricated on a stack consisting of five cycles of graphene/polymer layers [8]. The resonance frequency can be tuned by varying the disk diameter. (b) Extinction spectra of graphene nanoribbon arrays with a nanoribbon width of 150 nm and varying Fermi levels as indicated [10]. The solid vertical line indicates the energy of the CO_2 laser used in photocurrent experiments. (c) Gate-voltage-dependent photocurrent of a superlattice of 140-nm graphene nanoribbon arrays under incident infrared light with s-polarization (red spheres) and p-polarization (blue spheres) [10]. The corresponding Fermi energies are indicated on top. Inset: scanning photocurrent image of the graphene nanoribbon superlattice photodetector. Scale bar: 30 μm.

We found that localized plasmons can significantly enhance light absorption in 2D materials [8, 9]. Figure 3a shows the extinction spectra of graphene nanodisk arrays fabricated on a stack comprising five cycles of graphene and organic buffer layers [8]. We achieved a peak extinction of 85%, corresponding to a notch filter with an extinction ratio of approximately 8.2 dB [8]. The plasmon resonate frequency can be effectively controlled by adjusting the nanodisk diameter or nanoribbon width [8, 9]. t can also be dynamically tuned through electrostatic gating, thereby offering greater operational flexibility than traditional metal-based plasmonic structures. [10]. Figure 3b shows the resonance frequency position of the plasmon mode as a function of gate-voltage [10]. The plasmon resonance frequency is tuned by 75 cm^{-1}, when the Fermi level changes by 0.21 eV. These localized plasmons can significantly enhance the photocurrent. Figure 3c shows the photocurrent of a superlattice of 140 nm graphene nanoribbons under incident infrared light with s-polarization (along the nanoribbon width) and p-polarization (along the nanoribbon length) [10]. The photocurrent under s-polarization is more than an order of magnitude higher than under p-polarization. Here, localized plasmons are formed under s-polarization

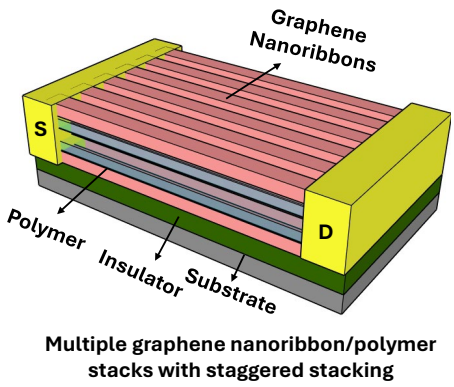

Figure 4. Proposed plasmon-enhanced photodetector based on multiple layers of graphene/polymer with staggered stacking.

due to the lateral confinement of the nanoribbons, while they are absent under p-polarization. The coupling of photons to the hybrid plasmon–phonon quasiparticles effectively enhances light absorption and leads to a significantly higher photocurrent.

In this project, we will explore plasmon-enhanced photodetectors based on multiple layers of 2D nanoribbons separated by polymer layers, as illustrated in Fig. 4. The 2D materials will be patterned into nanoribbon arrays, with ribbon width and spacing engineered to match the plasmon frequency to the target frequency. Multiple cycles of graphene and polymer layers are stacked to further enhance light absorption. In addition, to fully utilize the incident light, the graphene ribbons on odd layers are shifted by 50% of the ribbon period relative to those on even layers, so that the nanoribbons in the even layers lie directly beneath the gaps of the odd-layer ribbons. The etching of graphene ribbons will be optimized to minimize line edge roughness and reduce plasmon damping. Device structures and fabrication processes will be refined to realize ultrafast, highly sensitive photodetectors.

3. Exploration of Flexible Light Emitting Diodes

As wearable devices, foldable and rollable displays, biomedical sensors, and conformal lighting continue to expand, there is a growing demand for light sources that maintain high efficiency and reliability under mechanical deformation. Developing flexible LEDs enables new device form factors, enhances user comfort and durability, and allows seamless integration of light into soft, lightweight, and curved substrates. These advantages have driven intense research into flexible LEDs as a key building block for next-generation flexible and wearable optoelectronic systems.

Current flexible LED technologies mainly rely on rigid III–V semiconductor micro-LEDs or organic LEDs (OLEDs) integrated onto polymer substrates. While this approach enables bendability, it still faces limitations such as complex transfer processes, limited mechanical reliability under repeated bending, heat dissipation challenges, and constrained device form factors. Furthermore, device thickness and interfacial strain remain significant because the light-emitting layers are inherently bulk materials.

The PI has developed flexible RF devices based on graphene on polyimide, shown in Figure 5a. These devices exhibit excellent mechanical flexibility, sustaining bending down to a 1 mm radius [11]. In addition, the PI's group has successfully synthesized various TMD materials, including monolayer MoS_2, WSe_2, WS_2, $WSe_{2-2x}Te_{2x}$, and $MoTe_2$, with direct band gaps ranging from 1.1 to 2.1 eV [1-4, 7, 12], providing a wide spectral range for LEDs. Furthermore, we have synthesized monolayer $WSe_{2-2x}Te_{2x}$ with spatially graded compositions, core-shell structures, and lateral superlattices, all of which feature spatially controlled band gaps [2, 3, 12]. Figure 5b

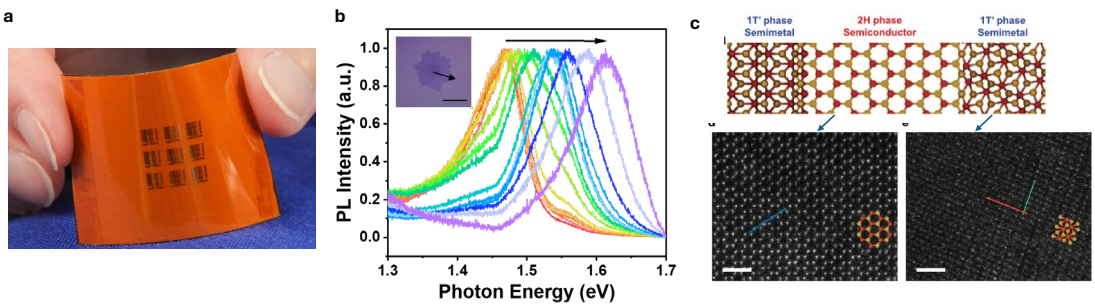

Figure 5. Preliminary Results on 2D material synthesis and flexible devices. (a) Optical image of arrays of graphene RF device on a flexible polyimide substrate [11]. (b) PL line-scanning spectra of the grown spatially graded $WSe_{2_2x}Te_{2x}$ flakes from the center to edge exhibit energy ranging from 1.46 to 1.61 eV [12]. Inset: the optical image of measured flake. Scale bar, 20 μm. (c) Atomic illustration of in-plane 1T–2H $MoTe_2$ heterophase junctions and measured high-magnification HAADF-STEM images of 2H and 1T′ $MoTe_2$ [7].

show the photoluminescence (PL) line-scanning spectra of the CVD $WSe_{2_2x}Te_{2x}$ with spatially graded compositions [12]. The laser scanning direction is indicated in the inset of Fig. 5b. The PL energy increases from 1.46 to 1.61 eV as the laser scans from the center to the edge. These tunable band gaps provide an effective means to engineer the emission wavelength in LEDs.

Moreover, we recently developed the synthesis of in-plane 2H–1T' heterophase structures, in which the semiconducting 2H and semi-metallic 1T' phases of $MoTe_2$ can be grown in a single step and seamlessly integrated at the atomic scale [7]. Figure 5c shows an atomic illustration of the $MoTe_2$ heterophase structure [7]. The high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images of the 2H and 1T' $MoTe_2$ regions are shown in Fig. 5c, overlaid with schematics of their in-plane atomic structures [7]. The hexagonal and rectangular lattices in the 2H and 1T' regions confirm the lateral heterophase structure. These semi-metallic 1T' TMDs can serve as highly conductive and flexible electrodes for LEDs.

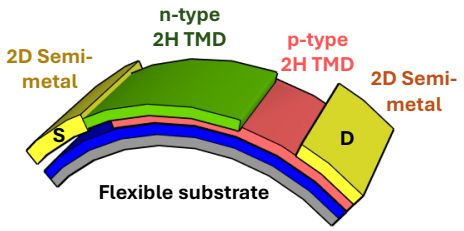

Figure 6. Schematic of a proposed flexible LED based on 2D materials.

In this task, we propose to develop flexible LEDs using 2H phase semiconducting TMDs as active light-emitting layer and 1T' phase semi-metallic TMDs as electrodes. This approach offers atomic-scale thickness, intrinsic flexibility, and van der Waals integration without lattice-matching constraints, potentially enabling ultrathin, mechanically robust, low-power flexible LEDs with new functionalities beyond conventional devices. The device schematic is illustrated in Fig. 6. A TMD p-n heterojunction, consisting of an n-type TMD monolayer (e.g., MoS_2 or $MoSe_2$) and a p-type TMD monolayer (e.g., WSe_2 or $MoTe_2$), will be fabricated on a flexible substrate. Monolayer TMDs have direct band gaps, enabling efficient electroluminescence. Their band gaps range from 1.1 eV ($MoTe_2$) to 2.1 eV (WS_2) [13, 14], corresponding to light emission wavelengths from 1.13 μm to 590 nm. To prevent mechanical failure of the metal electrode, 2D semi-metals, including graphene and semi-metallic 1T' WTe_2, will be used as the electrodes. Furthermore, multiple cycles of TMD p–n junctions, separated by polymer layers, will be employed to enhance the intensity of the emitted light.

R&D Target Specifications

The primary objective of this R&D project is to develop high-performance nanoscale electronic and optoelectronic devices with well-defined, measurable targets.

For logic transistors, the project aims to achieve:

- Drain current at on state: > 300 μA/μm;
- Drain-induced barrier lowering (DIBL): < 50 mV/V at 10 nm channel length;
- On/off current ratio: > 10^6;
- Subthreshold swing: < 80 mV/decade at 10 nm channel length.

For plasmon-enhanced photodetectors, the targets include:

- Response time: <1 ns;
- Frequency range: From 100 μm to 620 nm (from far-IR to visible);
- Responsivity: > 1 mA/W.

For flexible LEDs, the specifications are:

- Minimum bending radius: < 0.5 mm.
- Luminance: 100 cd/m²

	The project will evaluate these parameters under controlled conditions and benchmark performance against state-of-the-art devices.
R&D Plan	**Research Approaches:** In this project, we will pursue the following three research thrusts. 1. Demonstrate High-Performance Logic Transistors Based on Silicon Nanosheets and 2D Semiconductors In Thrust 1, we will develop multi-channel transistors using ultrathin silicon nanosheets and 2D transition metal dichalcogenides. For silicon nanosheet transistors, we will investigate precise control of the thickness and width of stacked silicon nanosheets using top-down etching, gate-all-around high-k/metal gate integration, and interface engineering to reduce variability and contact resistance. Device performance, reliability, and scalability will be systematically evaluated to establish design guidelines for next-generation, low-power, high-performance CMOS technologies. For 2D nanosheet transistors, we will synthesize large-scale, high-quality transition TMDs, develop low-resistance contacts, optimize the etching process to suspend the 2D layers, and engineer conformal deposition of insulator and metal layers surrounding the suspended 2D channels. We will characterize the performance of the logic transistor, including channel mobility, contact resistance, on/off drain current, and subthreshold swing. We will also study the impact of channel length scaling on the performance and reliability of the multi-channel transistors. 2. Develop Plasmon-Enhanced Photodetectors In Thrust 2, we will develop plasmon-enhanced photodetectors based on 2D nanostructures, aiming to achieve high speed, high sensitivity, and broad spectral response. In particular, we will carry out the following tasks: (1) fabricate arrays of 2D nanostructures, including nanodisks and nanoribbons, and characterize their plasmon resonant frequency and light absorption rate; (2) fabricate plasmon-enhanced photodetectors and evaluate their spectral photocurrent, responsivity, response time, external and internal quantum efficiency, and detectivity; (3) study the impact of geometry and carrier density of 2D nanostructures on photodetector performance; and (4) explore approaches for additional light absorption enhancement, including photonic cavities and anti-reflection films. 3. Investigate Flexible LEDs In Thrust 3, we will explore flexible LEDs based on semiconducting and semi-metallic 2D materials, aiming to achieve high flexibility and strong luminance. Specifically, we will carry out the following tasks. (1) Synthesize n-type TMDs, such as MoS_2 or $MoSe_2$, and p-type TMDs, such as WSe_2 and $MoTe_2$, using CVD or MOCVD. The n-type and p-type TMDs will be transferred and stacked vertically to form p-n heterojunctions. We will investigate the impact of growth conditions, including temperature, gas flow, and chamber pressure, on the quality and uniformity of the TMDs. The composition, band gap, film thickness, grain size, and defect density of the synthesized TMDs will be characterized. (2) Fabricate TMD LEDs on rigid substrates and characterize the electroluminescence spectrum under various bias conditions. We will investigate the impact of metal contacts, interface quality at the p-n heterojunctions, and carrier mobilities in the TMD layers on luminous efficacy and power consumption. (3) Develop TMD LEDs on flexible substrates. We will evaluate various flexible substrates, including polyimide, PET (polyethylene terephthalate), and PEN (polyethylene naphthalate). The light emission intensity and spectrum will be measured under different bending conditions, and the minimum bending radius before device failure will be benchmarked.

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	Milestones: • Milestone 1: Demonstrate multi-channel logic transistors with high drive current, low DIBL, and steep subthreshold swing. • Milestone 2: Gain insights into photon–plasmon interactions in 2D nanostructures. • Milestone 3: Demonstrate plasmon-enhanced photodetectors with ultrafast response and high sensitivity. • Milestone 4: Synthesize high-quality n-type and p-type TMD materials. • Milestone 5: Develop flexible LEDs with high luminance and small bending radius. **Timeline:** • Year 1: Develop multi-channel logic transistors based on silicon nanosheets and 2D semiconductors. • Year 2: Investigate localized plasmons in 2D nanostructures and develop plasmon-enhanced photodetectors. • Year 3: Synthesize 2H- and 1T'-phase 2D TMDs and develop flexible LEDs.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will serve as the primary driver of the proposed R&D project, providing expertise in material synthesis, device fabrication, and characterization. Its key roles and responsibilities include: 1. **Technical Contributions** • Synthesis of high-quality semiconducting and semi-metallic 2D materials; • Design of multi-channel logic transistors, plasmon-enhanced photodetectors, and flexible LEDs. • Optimization of device fabrication and processes integration. • Perform material characterization and device testing. 2. **Research Scope** • Conduct fundamental studies on charge transport in suspended 2D layers, photon-plasmon interactions, and the effect of strain on light emission. • Investigate the scalability and performance limit in logic transistors based on 2D nanosheet structures. • Develop plasmon-enhanced photodetectors aiming to achieve high responsivity and fast response time. • Explore flexible LEDs based on 2D materials, aiming to achieve mechanically robust devices with high luminance. 3. **Resources to Be Provided** • Access to cleanroom facilities, nanofabrication equipment, and optical/electrical characterization tools. • Skilled personnel including graduate students, engineers and technical staff. 4. **Specific Tasks** • Fabricate and characterize multi-channel logic transistors, plasmon-enhanced photodetectors, and flexible LEDs according to target specifications. • Analyze data, identify performance-limiting factors, and propose improvements. • Prepare reports, presentations, and publications documenting project outcomes.

	Through these roles, the institute ensures both scientific rigor and practical progress toward the project's objectives.
Expected R&R of Korean SME Partner	The Korean SME partner is expected to provide complementary expertise and capabilities that enable the translation of fundamental device research into application-ready technologies. Their key roles and responsibilities include: 1. **Technical Contributions** • Advise on device design, process, and integration. • Support optimization of material synthesis and device fabrication methods for manufacturability. 2. **Development Tasks** • Collaborate on prototype development for multi-channel logic transistors, plasmon-enhanced photodetectors, and flexible LEDs. • Implement design modifications to meet target performance specifications. 3. **Testing and Validation Activities** • Conduct device-level testing under industrial-relevant conditions. • Provide feedback on device performance and reliability. 4. **Collaboration Efforts** • Participate in regular project meetings and data reviews. • Contribute to joint reporting, presentations, and knowledge transfer. Through these contributions, the SME partner accelerates technology maturation and ensures alignment with industrial standards and application requirements.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The expected outcomes of this joint R&D project include both technological advances and broader industrial relevance. Technologically, the project aims to develop high-performance logic transistors with multiple channels, plasmon-enhanced photodetectors, and flexible LEDs, which can be integrated seamlessly in a single system. Prototype devices will demonstrate target performance metrics, including high drive current, steep subthreshold slope, low DIBL, fast photo response times, efficient light emission, and excellent mechanical flexibility, establishing a foundation for future integrated electronic–optoelectronic systems. These devices will have high potential in advanced computing, optical communications, advanced displays, and bio-medical applications. From a commercialization perspective, the resulting device platform is directly relevant to semiconductor, communication, and display industries. The collaboration with the SME partner ensures that fabrication processes, material selection, and device architectures are aligned with practical manufacturing constraints, increasing the likelihood of industrial adoption. The broader industrial and economic impacts of this project include accelerating the development of next-generation computing, optical communication, and display technologies. By enabling compact, flexible, and high-performance devices with multiple functionalities, the project contributes to energy-efficient electronics, supports advanced manufacturing capabilities, and strengthens the competitive position of both academic and industrial partners in the global semiconductor and optoelectronic markets.

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6. References

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